MILITARY SPECIFICATION

PROPELLANT M30
FOR USE IN 105MM CARTRIDGE, HEAT-T, M622

This specification is approved for use by the US Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope This specification covers the requirements, quality assurance provisions, and packaging of M30 Propellant for use in the 105MM Cartridge HEAT-T M622 for the M102, M101A1 and M204, Howitzer.

1.2 Classification. The propellant shall be Type 1, cylindrical multiperforated grains.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and Standards. Unless otherwise specified (see 6.1), the following specifications and standards of issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for

FSC: 1376

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-414 - Sampling Procedures and Tables For Inspection By Variables For Percent Defective

MIL-STD-652 - Propellant, Solid, for Cannons, Requirements and Packing

2.1.2 Other government documents, drawings and publications. The following documents and drawings form a part of this specification to the extent specified herein.

DRAWINGS (see 6.9)

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

9282517 - Cartridge, 105MM, HEAT-T, M622
9282518 - Projectile, 105MM, HEAT-T, M622 Loading, Assembly,

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Order of precedence In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

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3. REQUIREMENTS

3.1 Material. The propellant shall be M30 Type 1, conforming to the requirements of MIL-STD-652 and with a nominal web of 0.033 inch (in.).

3.2 Ballistic assessment requirements. The propellant when fired in the M102 Howitzer with the weight of projectile (M622, 22.75 lbs + .02 lbs) shall comply with the following requirements at 70 degrees Fahrenheit (^{o}F) (see 4.5.1) and standard weapon conditions (see 6.11). The propellant shall be capable of assessment of a velocity of 2270 feet per second (fps) with an average pressure not to exceed 39,000 pounds per square inch (psi) and a charge weight not to exceed 55 ounces (oz) as determined in 4.5.1.1.

3.3 Ballistic uniformity requirements. The propellant when fired in the M102 Howitzer with the projectile M622 shall comply with the following requirements when tested as specified in 4.5.1.2.

3.3.1 Uniformity of velocity. The standard deviation of velocity for the lot at 70°F shall not exceed 8.0 fps.

3.3.2 Average Pressure. At 70°F and standard weapon conditions, the propellant shall not produce an average pressure greater than 39,000 psi.

3.3.3 Individual maximum pressure. At 145°F and standard weapon conditions (see 6.11), the propellant shall not produce an individual pressure greater than 45,600 psi or an average pressure plus three (3) standard deviations greater than 45,600 psi.

3.4 Preproduction sample inspection. This specification contains provisions for preproduction sample inspection. Requirements for the submission of preproduction samples shall be as specified by the contracting officer.

3.5 Workmanship. The best commercial practices shall be used in the formulation of propellant furnished under this specification, and all other applicable documents. The propellant and its standard ingredients shall be protected from the action of direct sunlight and acid fumes. Unless otherwise specified, no reground propellant or nitrocellulose shall be used in the manufacture of propellant procured under this specification.

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4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections. The following types of inspection shall be conducted on the unit of product:

a. Preproduction sample inspection

b. Quality conformance inspection

4.3 Preproduction sample inspection.

4.3.1 Submission. The contractor shall select three (3) propellant granulations which, from experience, are expected to bracket the requirements of this specification. The contractor shall submit a preproduction sample as designated by the Contracting Officer for evaluation in accordance with 4.3.2, consisting of three (3) pilot lots of 500 lbs to 1000 lbs each representing the three (3) candidate granulations.

4.3.2 Inspections to be performed. Each of the pilot lots comprising the preproduction sample and submitted in accordance with 4.3.1 may be subjected by the Government to any or all of the requirements of the applicable drawings and specifications.

4.3.3 Rejection. Government Engineering and Product Assurance personnel and the Contractor shall review the pilot lot test data and shall select one granulation which best fits the requirements of this specification. The Technical Agency shall notify the Contracting officer of the selected granulation. The Contractor shall request approval from the contracting officer to produce the selected granulation. If none of the pilot lots submitted meets the requirements of this specification, a new preproduction sample shall be submitted in accordance with 4.3.1. Until the preproduction sample is accepted, the Contractor is in no way authorized by the Government to initiate regular production, unless otherwise directed by the Contracting Officer. All future production lots should be of the same basic granulation and process as the accepted pilot lot.

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4.4 Quality conformance inspections

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078.

PRECAUTION - This specification covers sampling and testing of chemical, toxic or explosive materials which are potentially hazardous to personnel. Accordingly, it is emphasized that all applicable safety rules, regulations and procedures must be followed in handling and processing these materials.

4.4.2 Examination and testing

4.4.2.1 Sampling for laboratory testing (see 3.1). Each lot of propellant shall be sampled and tested for composition, form, dimensions and chemical/physical properties in accordance with MIL-STD-652.

4.4.2.2 Sampling for ballistic testing (see 3.2). Each lot of propellant shall be sampled for Proving Ground tests in accordance with MIL-STD-652.

4.4.2.3 Charge establishment (see 3.2). If the lot does not meet the applicable requirements, testing shall cease and the Test Director shall immediately notify the contracting officer.

4.4.2.4 Velocity uniformity (70°F) (see 3.3.1). If the product of the factor (0.73) times the velocity standard deviation of the rounds tested in accordance with 4.5.1.2 exceeds the applicable requirement, apply retest and referral criteria in accordance with 4.5.1.3 and 4.5.1.4. If the retest and referral criteria cited do not apply, the lot shall be rejected.

4.4.2.5 Average pressure (see 3.3.2) If any average pressure obtained during initial or subsequent production at 70°F, when corrected to standard projectile weight, standard conditions and recommended charge weight, exceeds the applicable requirement, apply retest and referral criteria in accordance with 4.5.1.3 and 4.5.1.4. If the retest and referral criteria do not apply, the lot shall be rejected.

4.4.2.6 Individual maximum pressure (IMP) (see 3.3.3).

4.4.2.6.1 If any individual pressure observed during the uniformity series, when corrected to standard projectile weight and standard conditions and recommended charge weight exceeds the applicable requirement, the lot shall be rejected.

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4.4.2.6.2 If the average pressure at 145°F plus three standard deviations exceeds the applicable requirement, apply referral criteria in accordance with 4.5.1.4. If the referral criteria cited do not apply, the lot shall be rejected.

4.5 Test methods and procedures

4.5.1 Proving ground assessments. These tests shall be conducted at a Government Proving Ground (designated by the contracting officer) in accordance with the applicable Proving Ground Acceptance Test Procedure. Tubes used for these assessments shall be restricted to fire only High Explosive Anti-Tank with Tracer (HEAT-T) ammunition. The wear of the tube shall be no greater then 0.015 in when measured 15.50 in from the rear face of the tube. All test rounds shall be temperature conditioned in a range conditioning box, with adequate air circulation to assure uniform conditioning, at the required temperature for a minimum of 24 hours and assurance will be made that when fired, the rounds are at the conditioning temperature. The following tolerances shall apply to all temperature conditionings: -50°F +5°F; 70°F, ±2.5°F; 145°F, -5°F. Two copper crusher gages shall be used to measure the maximum chamber pressure. Velocity coils shall be used to measure the muzzle velocity. Test rounds shall be assembled in accordance with drawing listed in 2.1.

4.5.1.1 Charge establishment.

4.5.1.1.1 Charge slope method. Three or more groups of rounds, three rounds for each group loaded with selected charge weights of propellant (weights selected should bracket expected service charge weight) shall be conditioned and fired at 70°F. Three calibration rounds shall also be conditioned and fired at 70°F prior to the test rounds. From the data obtained, a composite charge weight versus velocity versus pressure curve shall be established. A charge weight shall be calculated which will yield the prescribed service velocity (see 3.2). This calculated charge weight shall then be loaded into rounds for the uniformity series (see 4.5.1.2.1). The difference between the corrected velocity obtained during the uniformity series and the service velocity shall be compensated for by a final adjustment in charge weight using the slope of the velocity charge weight curve previously established. The resultant will be the recommended charge weight.

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4.5.1.1.2 Constant slope method. After ten (10) lots of a given production series have been fired to provide a good knowledge of the propellant characteristics, the charge slope method of assessment may be discontinued and the charge weight determined by the constant slope method. A composite charge weight versus velocity versus pressure curve can be established from test results generated from the ten lots. This curve can then be used as in 4.5.1.1.1 to calculate the charge weight needed to achieve service velocity. This calculated charge weight is then to be loaded into rounds for the uniformity series and final adjustment is to be made as indicated above for the charge slope method (see 4.5.1.1.1). The resultant will be the recommended charge weight.

4.5.1.2 Uniformity.

4.5.1.2.1 Initial production. Thirty (30) rounds shall be assembled with the established charge weight (see 4.5.1.1.1) of propellant. Ten rounds each shall be conditioned to -50°F, 70°F and 145°F (as indicated by thermocoupled control round). Rounds fired at 70°F shall be fired alternately with calibration rounds and shall be corrected to standard weapon/firing conditions (i.e., the difference between calibration performance under test conditions and standard conditions). Firing results at all temperatures shall be corrected to standard projectile weight (see 3.2) and to recommended charge weight conditions. The means and standard deviations of the velocities and pressures at all firing temperatures shall be calculated using corrected results. Testing at -50°F shall be for information only. The initial production lots represent the first ten consecutive lots from each manufacturer which have been produced and met the acceptance criteria of this specification.

4.5.1.2.2 Subsequent production. Twenty (20) rounds shall be assembled with the established charge weight (see 4.5.1.1.2) of propellant. Ten rounds each shall be conditioned to 70°F and 145°F (as indicated by thermocoupled control round). Rounds fired at 70°F shall be fired alternately with calibration rounds and shall be corrected to standard weapon/firing conditions (i.e., the difference between calibraton performance under test conditions and standard conditions). Firing results at both temperatures shall be corrected to standard projectile weight (see 3.2). The means and standard deviations of the velocities and pressures at all firing temperatures shall be calculated using corrected results. Throughout production, if 3 consecutive lots fail on first test, testing shall revert back to the initial production plan (see 4.5.1.2.1).

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4.5.1.3 Retest criteria. A lot shall be retested whenever any of the following conditions exist and whenever the proving ground deems it necessary.

a. The final adjustment in charge weight exceeds two ounces of the charge weight used in uniformity firings, retest for charge establishment and uniformity at 70°F.

b. If for any reason, the proving ground considers that the test conditions have detrimentally affected the test results, additional rounds as required shall be tested.

4.5.1.4 Referral criteria. A test lot shall be referred to the contracting officer for disposition whenever any of the following conditions exist and whenever the proving ground deems it necessary.

a. Calibration rounds fired in any test phase fail to comply with Section 3 of this specification.

b. Calibration correction applied exceeds 25 fps.

c. Calibration components are not available.

d. Number of test or calibration results utilized in calculations is less than specified.

e. Any unusual occurrences during ballistic testing, such as excessive delays between rounds, stickers, or projectiles flights, etc. Each such event shall be reported in detail on the applicable firing record.

f. Test lot fails velocity standard deviation or average pressure requirement at 70°F and any extreme test value is an outlier.

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The following test shall be performed to determine if a single test result is an outlier with a sample size of 10 at a significance level of 0.05, the critical value (T_C) is 0.477. Calculate a test value (T_t) as follows:

For Low Outlier

$$T_t = \frac{(X_2 - X_1)}{(X_9 - X_1)}$$

For High Outlier

$$T_t = \frac{(X_{10} - X_9)}{(X_{10} - X_2)}$$

Where:

X_1 = test result being tested as low outlier
X_2 = test result adjacent to X_1, after all the test results are ordered from smallest to largest
X_9 = test result adjacent to the largest test result
X_{10} = test result being tested as high outlier

X_1 or X_{10} is an outlier if T_t is greater than T_C.

5. PACKAGING

5. Packing and marking. Packing and marking shall be in accordance with MIL-STD-652, except that all propellant designated for use as reference shall be packed Level A.

6. NOTES

6.1 Ordering data. Ordering data shall be in accordance with MIL-A-48078 (PA) with the following exceptions.

6.1.1 Acquisition requirements. Acquisition requirements of MIL-A-48078 (PA) shall apply.

6.1.2 Contract data requirements. In addition to the normal distribution of records, when the propellant is procured by the Department of the Army, one copy of all ballistic test data and one copy of the Propellant Description sheet shall be forwarded to: Commander U. S. Army R&D Commmand, ATTN : DRDAR-QAR-R, and one copy to ATTN: DRDAR-LCA-G Dover, NJ 07801.

6.2 Standard deviation. The standard deviation shall be calculated with (n-1) as the divisor in a standard statistical technique equivalent to that shown in MIL-STD-414, Section B, example B-1.

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6.3 Factors. The factors, as taken from Table A-21 of AMC Pamphlet 706-114 using the 95 percent upper confidence limits (A.05) and N-1 degrees of freedom, make allowances of the probability that a standard deviation from a smaller sample size exceeds the true standard deviation by chance alone.

6.4 Surveillance sample. Samples shall be forwarded to U.S. Army R&D Command, Dover, NJ, Attn: DRDAR-LCE-MP for the propellant surveillance test program in accordance with MIL-STD-652.

6.5 Test rounds. Test rounds shall be assembled in accordance with drawing 9282517, 9282518.

6.6 Proving ground test summary

Initial Production

Temp (F)	Sample size	Assessed velocity (fps)	Lot Std Dev of Vel (fps)	Ind Max Press (psi)	Avg Press (psi)
-50	10	N/A	N/A	N/A	N/A
70	10	2270	8.0	-	39,000
145	10	N/A	N/A	45,600	-

Subsequent production

70	10	2270	8.0	-	39,000
145	10	N/A	N/A	45,600	-

6.7 Retest. All original and retest data shall be submitted to U.S. Army R&D Command, ATTN: DRDAR-QAR-R, Dover, NJ 07801.

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6.8 Test anomalies. Any unusual occurrences during ballistic testing, such as excessive delays between rounds, stickers, poor projectile flights, etc., shall be reported in detail on the applicable firing record.

6.9 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Armament Research and Development Command (ARRADOCM) may also include drawings prepared by, and identified as, Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARRADCOM.

6.10 Individual maximum pressure The individual maximum pressure value for 145°F cited is based upon the weapon design pressure.

6.11 Standard weapon Standard weapon and standard weapon conditions refer to the weapon model and conditions of the test at the time the master calibration series was fired.

6.12 Preproduction sample. The preproduction sample is intended to provide the contractor with the necessary latitude to establish the propellant granulation needed to meet the required ballistic parameters. the Technical Agency and the contractor should jointly select one granulation which best fits the requirements. All future production lots should be of the same basic granulation as the accepted granulation.

6.13 Reference sample. Whenever a propellant lot is selected for reference use, a 5- lb sample shall be forwarded to US Army Armament R&D Command, Dover, NJ 07801, ATTN: DRDAR-LCA-G for use in evaluating the propellant stockpile.

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